

GOSFORD HIGH SCHOOL



Year 12

2008

Higher School Certificate

MATHEMATICS EXTENSION I

Half Yearly Examination

Time Allowed: 2 Hours + 5 minutes reading time

Instructions:

- Start each question in a new booklet (extra booklets are available).
- Attempt questions 1-7.
- All questions are of equal value.
- Board approved calculators may be used.
- Write using black or blue pen.
- A table of standard integrals is provided at the back of this paper.
- All necessary working should be shown in every question.

QUESTION 1: (12 Marks) Use a separate writing booklet.**Marks**

- a. Expand and simplify

$$(3x - 5)(2x + 1) - (2x + 3)^2$$

2

- b. Factorise completely

$$x^2z + xyz + y^2z - x^3 + y^3$$

2

- c. Differentiate with respect to
- x

$$(5x + 1)^3(x - 9)^5$$

2

Expressing your answer in a form that is fully factorised.

- d. Divide
- $P(x) = 3x^5 - 7x^3 + 8x^2 - 5$
-
- by
- $x - 2$
- and write
- $P(x)$
- in the
-
- form
- $P(x) = (x - 2)Q(x) + R(x)$

3

- e. If
- α, β, γ
- are the roots of

$$x^3 - 2x^2 + 3x - 5 = 0$$

3

Evaluate:

i. $\alpha + \beta + \gamma$

ii. $\alpha\beta\gamma$

iii. $\frac{1}{\alpha} + \frac{1}{\beta} + \frac{1}{\gamma}$

QUESTION 2: (12 Marks) Use a separate writing booklet.

	Marks
a. Differentiate $\sin^3 x$ with respect to x .	1
b. Find $\int 4 \sec^2\left(\frac{x}{2}\right) dx$	1
c. Solve $2 \cos^2 x - \sqrt{3} \cos x = 0$ for $0 \leq x \leq 2\pi$, writing your answers in exact radian form.	2
d. i. Express $6 \sin x + 8 \cos x$ in the form $A \sin(x + \theta)$ where $A > 0$, $0 \leq \theta \leq \frac{\pi}{2}$	1
ii. Hence, solve $6 \sin x + 8 \cos x = 5$ for $0 \leq x \leq 2\pi$	2
(Answers in radians correct to 2 decimal places)	
e. Show that $\frac{\cos \theta}{\sin \theta + 1} = \frac{1-t}{1+t}$ where $t = \tan \frac{\theta}{2}$	2
f. The area under the curve $y = \sin x + \cos x$ above the x axis and between $x = 0$ and $x = \frac{\pi}{2}$ is rotated about the x axis. Find the volume of the solid of revolution formed. (<i>Leave answer in exact form</i>)	3

QUESTION 3: (12 Marks) Use a separate writing booklet.**Marks**

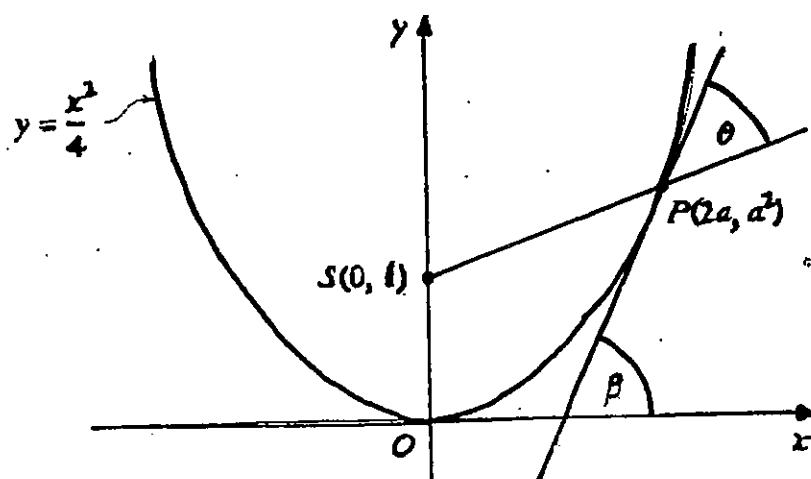
a. Two points $P(2ap, ap^2)$ and $Q(2aq, aq^2)$ lie on the parabola $x^2 = 4ay$

i. Show that the equation of the tangent to parabola at P is $y = px - ap^2$ 1

ii. The tangent at P and the line through Q parallel to the y axis intersect at T. Find the co-ordinates of T. 1

iii. Write down the co-ordinates of M, the midpoint of PT 1

iv. Describe the locus of M when $pq = -1$ 1



Let $P(2a, a^2)$ be a point on the parabola $y = \frac{x^2}{4}$
and let S be the point $(0, 1)$.

The tangent to the parabola at P makes an angle of β
with the x axis. The angle between SP and the tangent is θ .

Assume that $a > 0$.

Show that

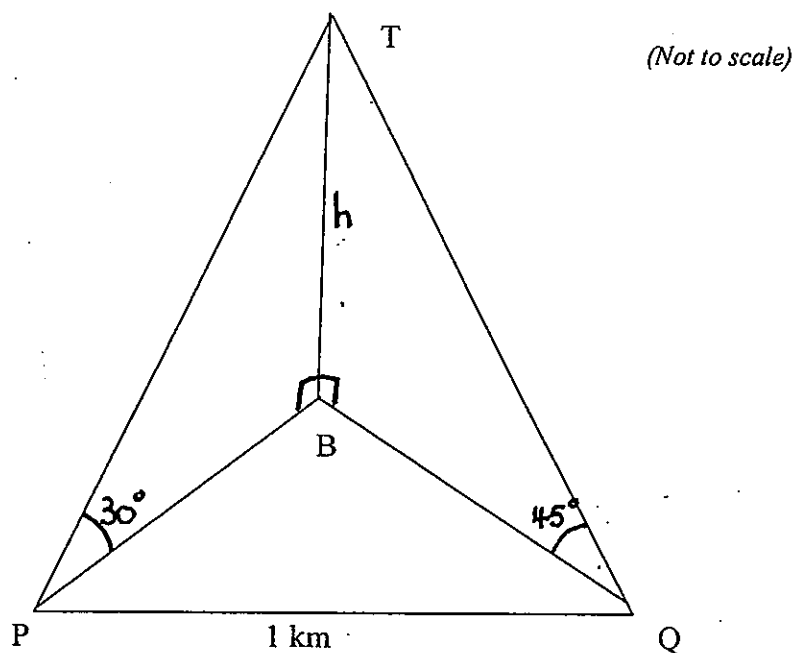
i. $\tan \beta = a$ 1

ii. the gradient of SP is $\frac{1}{2}(a - \frac{1}{a})$ 1

iii. $\tan \theta = \frac{1}{a}$ 2

Question 3 (Continued)

c.



The angle of elevation from a boat P to a point T at the top of a vertical cliff is measured to be 30° . The boat sails 1 km to a second point Q, from which the angle of elevation of T is 45° . Let B be the point at the base of the cliff directly below T and let $h = BT$, the height of the cliff in metres. The bearings of B from P and Q are 50° and 310° respectively.

- i. Show that $\angle PBQ = 100^\circ$ 1
- ii. Find expression for PB and QB in terms of h 1
- iii. Hence, show that 1

$$h^2 = \frac{1000^2}{\cot^2 30^\circ + \cot^2 45^\circ - 2 \cot 30^\circ \cot 45^\circ \cos 100^\circ}$$
- iv. Use a calculator to find the height of the cliff, correct to the nearest metre. 1

QUESTION 4: (12 Marks) Use a separate writing booklet.

	Marks
a. Given that	
$\frac{1}{1 \times 4} + \frac{1}{4 \times 7} + \dots + \frac{1}{(3k-2)(3k+1)} = \frac{k}{3k+1}$	2
prove that	
$\frac{1}{1 \times 4} + \frac{1}{4 \times 7} + \dots + \frac{1}{(3k-2)(3k+1)} + \frac{1}{(3k+1)(3k+4)} = \frac{k+1}{3k+4}$	
b. Use the principle of Mathematical Induction to prove that	
$2^{3n} - 3^n$ is divisible by 5	4
for all positive integers n	
c. i. Prove that $f(x) = x^3 + 3x - 18$ has a root in the interval $2 < x < 2.5$	1
ii. Use one application of the 'halving the interval' method to find a smaller interval containing the root.	1
iii. Which end of the small interval found in (ii) is closer to the root. Justify your answer.	1
d. Given that 9.5 is an approximate root of $x^3 - 6x^2 + 25x - 500 = 0$, find by Newton's Method a better approximation. (Give your answer to 2 decimal places.)	3

QUESTION 5: (12 Marks) Use a separate writing booklet.

		Marks
a.	Find:	
i.	$\int \frac{\cos x}{\sin x} dx$	1
ii.	$\int \sin^2 4x dx$	2
b.	Use the table of standard integrals to evaluate:	
	$\int_2^3 \frac{dx}{\sqrt{x^2 - 4}}$	2
c.	Evaluate:	
	$\int_0^\pi (1 + \cos x)^2 dx$	2
d.	Find:	
	$\int \frac{x}{\sqrt{1+x^2}} dx$	2
	Using the substitution $x = \tan \theta$	
e.	Evaluate	
	$\int_0^3 \frac{3x}{\sqrt{1+x}} dx$	3
	Using the substitution $u = 1+x$	

QUESTION 6: (12 Marks) Use a separate writing booklet.**Marks**

a. A tangent is drawn to $y = 2 \tan^{-1} x$ at point P where $x = 1$

i. Find the y co-ordinate of P.

1

ii. Find the gradient of the normal at P.

1

b. Evaluate

$$\int_0^{\sqrt{3}} \frac{4}{9+x^2} dx$$

2

c. Determine the exact value of

$$\cos\left(\sin^{-1}\left(\frac{-12}{13}\right)\right)$$

2

d. For $f(x) = x^2 - 6x + 6$

i. Find the co-ordinates of the vertex.

1

ii. Explain why $f(x)$ does not have an inverse function and state the restricted domain of $f(x)$ over which an inverse exists which contains all positive x values. Call this inverse function $y = f^{-1}(x)$

1

iii. State the domain of $f^{-1}(x)$

1

iv. Find an expression for $y = f^{-1}(x)$ in terms of x .

2

v. Find the x values for the points of intersection of $y = f(x)$ and $y = f^{-1}(x)$

1

QUESTION 7: (12 Marks) Use a separate writing booklet.

		Marks
a.	Solve $\frac{ x-1 -2}{12+x-x^2} \geq 0$	2
b.	A function $y = f(x)$ is defined by $f(x) = 2 \cos^{-1}\left(\frac{x}{\sqrt{2}}\right) - \sin^{-1}(1-x^2)$ and restricted to values of $x > 0$	
i.	Find the domain of $f(x)$	1
ii.	Find $f'(x)$	2
iii.	Hence, sketch $y = f(x)$	1
c.	Prove that: $\frac{\sin\left(\frac{3x}{2}\right)}{2\sin\left(\frac{x}{2}\right)} = \frac{1}{2} + \cos x$	3
d.	Given that the equation $P(x) = x^4 - 4kx^2 + 12$ has a double root, find all possible value(s) of k	3

END OF TEST

STANDARD INTEGRALS

$$\int x^n dx = \frac{1}{n+1} x^{n+1}, \quad n \neq -1; \quad x \neq 0, \text{ if } n < 0$$

$$\int \frac{1}{x} dx = \ln x, \quad x > 0$$

$$\int e^{ax} dx = \frac{1}{a} e^{ax}, \quad a \neq 0$$

$$\int \cos ax dx = \frac{1}{a} \sin ax, \quad a \neq 0$$

$$\int \sin ax dx = -\frac{1}{a} \cos ax, \quad a \neq 0$$

$$\int \sec^2 ax dx = \frac{1}{a} \tan ax, \quad a \neq 0$$

$$\int \sec ax \tan ax dx = \frac{1}{a} \sec ax, \quad a \neq 0$$

$$\int \frac{1}{a^2 + x^2} dx = \frac{1}{a} \tan^{-1} \frac{x}{a}, \quad a \neq 0$$

$$\int \frac{1}{\sqrt{a^2 - x^2}} dx = \sin^{-1} \frac{x}{a}, \quad a > 0, \quad -a < x < a$$

$$\int \frac{1}{\sqrt{x^2 - a^2}} dx = \ln \left(x + \sqrt{x^2 - a^2} \right), \quad x > a > 0$$

$$\int \frac{1}{\sqrt{x^2 + a^2}} dx = \ln \left(x + \sqrt{x^2 + a^2} \right)$$

NOTE : $\ln x = \log_e x, \quad x > 0$

$$\begin{aligned} a) & (3x-5)(2x+1) - (2x+3)^2 \\ &= 6x^2 + 3x - 10x - 5 - (4x^2 + 12x + 9) \\ &= 6x^2 - 7x - 5 - 4x^2 - 12x - 9 \\ &= 2x^2 - 19x - 14 \end{aligned}$$

$$\begin{aligned} b) & x^2z + xyz + y^2z - x^3 + y^3 \\ &= (x^2z + xyz + y^2z) - (x^3 - y^3) \\ &= z(x^2 + xy + y^2) - (x-y)(x^2 + xy + y^2) \\ &= (x^2 + xy + y^2)(z - x + y) \end{aligned}$$

$$\begin{aligned} c) & \text{Let } y = (5x+1)^3(x-9)^5 \\ y' &= (x-9)^5 \cdot 15(5x+1)^2 \\ &\quad + (5x+1)^3 \cdot 5(x-9)^4 \\ &= 5(5x+1)^2(x-9)^4 \left[(x-9) \cdot 3 + (5x+1) \right] \\ &= 5(5x+1)^2(x-9)^4 [3x - 27 + 5x + 1] \end{aligned}$$

$$\begin{aligned} &= 5(5x+1)^2(x-9)^4(8x-26) \\ &= 10(5x+1)^2(x-9)^4(4x-13) \end{aligned}$$

$$\begin{array}{r} d) \quad \begin{array}{r} 3x^4 + 6x^3 + 5x^2 + 18x + 36 \\ x-2 \overline{) 3x^5 - 7x^3 + 8x^2 - 5} \\ \underline{3x^5 - 6x^4} \\ 6x^4 - 7x^3 \\ \underline{6x^4 - 12x^3} \\ 5x^3 + 8x^2 \\ \underline{5x^3 - 10x^2} \\ 18x^2 - 5 \\ \underline{18x^2 - 36x} \\ 36x - 5 \\ \underline{36x - 72} \\ 67 \end{array} \end{array}$$

$$\therefore P(x) = (x-2)(3x^4 + 6x^3 + 5x^2 + 18x + 36) + 67$$

$$e) \quad \alpha + \beta + \gamma = -\frac{b}{a} = 2$$

$$\alpha\beta + \alpha\gamma + \beta\gamma = \frac{c}{a} = 3$$

$$\alpha\beta\gamma = -\frac{d}{a} = 5$$

$$(i) \quad 2$$

$$(ii) \quad 5$$

$$\begin{aligned} (iii) \quad \frac{1}{\alpha} + \frac{1}{\beta} + \frac{1}{\gamma} &= \frac{\beta\gamma + \alpha\gamma + \alpha\beta}{\alpha\beta\gamma} \\ &= \frac{3}{5} \end{aligned}$$

2/a) Let $y = \sin^3 x$

$$y' = 3 \sin^2 x \cdot \cos x$$

$$= 3 \cos x \sin^2 x$$

b) $\int 4 \sec^2\left(\frac{x}{2}\right) dx$

$$= 2 \cdot 4 \tan\left(\frac{x}{2}\right) + C$$

$$= 8 \tan\left(\frac{x}{2}\right) + C$$

c) $2 \cos^2 x - \sqrt{3} \cos x = 0$

$$\cos x (2 \cos x - \sqrt{3}) = 0$$

$$\therefore \cos x = 0, 2 \cos x - \sqrt{3} = 0$$

$$\cos x = \frac{\sqrt{3}}{2}$$

$$\therefore x = \frac{\pi}{2}, \frac{3\pi}{2}, \frac{\pi}{6}, \frac{11\pi}{6}$$

$$\therefore x = \frac{\pi}{6}, \frac{5\pi}{6}, \frac{7\pi}{6}, \frac{11\pi}{6}$$

d) (i) $6 \sin x + 8 \cos x = A \sin(x + \theta)$

$$A = \sqrt{6^2 + 8^2} = 10 \quad \tan \theta = \frac{8}{6}$$

$$\therefore \theta = \tan^{-1} \frac{4}{3}$$

$$\therefore 6 \sin x + 8 \cos x = 10 \sin\left(x + \tan^{-1}\left(\frac{4}{3}\right)\right)$$

(ii) $10 \sin\left(x + \tan^{-1}\frac{4}{3}\right) = 5$

$$\sin\left(x + \tan^{-1}\frac{4}{3}\right) = \frac{1}{2}$$

$$\therefore x + \tan^{-1}\frac{4}{3} = \frac{\pi}{6}, \frac{5\pi}{6}, \frac{13\pi}{6}, \dots$$

$$\therefore x + 0.927 = \frac{\pi}{6}, \frac{5\pi}{6}, \frac{13\pi}{6}$$

$$x = 1.69, 5.88 \text{ (2 d.p.)}$$

e) $\frac{\cos \theta}{\sin \theta + 1} = \frac{1-t^2}{1+t^2}$

$$\frac{2t}{1+t^2} + 1$$

$$= \frac{1-t^2}{1+t^2}$$

$$\frac{2t+1+t^2}{1+t^2}$$

$$= \frac{1-t^2}{(1+t)^2}$$

$$= \frac{(1-t)(1+t)}{(1+t)^2}$$

$$= \frac{1-t}{1+t} \text{ as req.}$$

f) $V = \pi \int_0^{\frac{\pi}{2}} (\sin x + \cos x)^2 dx$

$$= \pi \int_0^{\frac{\pi}{2}} \sin^2 x + 2 \sin x \cos x + \cos^2 x$$

$$= \pi \int_0^{\frac{\pi}{2}} 1 + 2 \sin x \cos x dx$$

$$= \pi \int_0^{\frac{\pi}{2}} 1 + \sin 2x \, dx$$

$$= \pi \left[x - \frac{1}{2} \cos 2x \right]_0^{\frac{\pi}{2}}$$

$$= \pi \left[\left(\frac{\pi}{2} - \frac{1}{2} \cos \pi \right) - \left(0 - \frac{1}{2} \cos 0 \right) \right]$$

$$= \pi \left[\left(\frac{\pi}{2} + \frac{1}{2} \right) - \left(-\frac{1}{2} \right) \right]$$

$$= \pi \left[\frac{\pi}{2} + 1 \right] \text{ units}^3$$

3/ a) (i) $x^2 = 4ay$

$$y = \frac{x^2}{4a}$$

$$y' = \frac{2x}{4a}$$

(when $x = 2ap$) $y' = \frac{2(2ap)}{4a}$
 $= p.$

$\therefore$ Eqn of tangent $\Rightarrow$

$$y - ap^2 = p(x - 2ap)$$

$$y - ap^2 = px - 2ap^2$$

$$y = px - ap^2 \text{ as req.}$$

(ii) line through Q: $x = 2aq$

Solving $y = px - ap^2$ — (1)

$$x = 2aq \quad \text{--- (2)}$$

Sub (2) in (1): $y = 2apq - ap^2$

$$\therefore T = (2aq, 2apq - ap^2)$$

(iii) $M = \left(\frac{2ap + 2aq}{2}, \frac{ap^2 + 2apq - ap^2}{2} \right)$

$$= (a(p+q), apq)$$

(iv) when $pq = -1$

$$M = (a(p+q), -a)$$

Since the y co-ordinate is $-a$
 the locus of M is the directrix

b) (i) $y' = \frac{2x}{4} = \frac{x}{2}$

(at $x = 2a$) $y' = \frac{2a}{2} = a$

$\therefore$ gradient of tangent $= a$

$\therefore \tan \beta = a$

(since $\tan \beta = m$)

(ii) gradient SP $= \frac{a^2 - 1}{2a}$

$$= \frac{a}{2} - \frac{1}{2a}$$

$$= \frac{1}{2} \left(a - \frac{1}{a} \right)$$

as req.

(iii) $\tan \theta = \frac{m_1 - m_2}{1 + m_1 m_2}$

$$\tan \theta = \frac{a - \frac{1}{2} \left(a - \frac{1}{a} \right)}{1 + \frac{a}{2} \left(a - \frac{1}{a} \right)}$$

$$\therefore \tan \theta = \frac{a - \frac{a}{2} + \frac{1}{2a}}{1 + \frac{a^2}{2} - \frac{1}{2}}$$

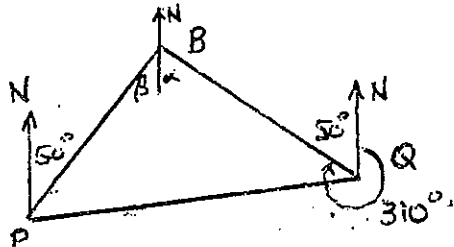
$$= \frac{\frac{a}{2} + \frac{1}{2a}}{\frac{a^2}{2} + \frac{1}{2}}$$

$$= \frac{a + \frac{1}{a}}{a^2 + 1}$$

$$= \frac{\frac{a^2 + 1}{a}}{a^2 + 1}$$

$$= \frac{1}{a} \text{ as req.}$$

c) (i) Consider:



From diagram: $\angle = 50^\circ$ (alternate $\angle$'s in $\parallel$ are $=$)
Similarly $\beta = 50^\circ$

$$\therefore \hat{P}BQ = 100^\circ \text{ (angle sum of } \triangle PBQ = 180^\circ)$$

$$(ii) \tan 30^\circ = \frac{h}{PB}$$

$$PB = \frac{h}{\tan 30^\circ}$$

$$= h \cot 30^\circ$$

$$\therefore QB = h \cot 45^\circ$$

(iii) using $\triangle PBQ$ (cosine rule)

$$1000^2 = h^2 \cot^2 30^\circ + h^2 \cot^2 45^\circ - 2h^2 \cot 30^\circ \cot 45^\circ \cos 100^\circ$$

$$1000^2 = h^2 [\cot^2 30^\circ + \cot^2 45^\circ - 2 \cot 30^\circ \cot 45^\circ \cos 100^\circ]$$

$$\therefore h^2 = \frac{1000^2}{\cot^2 30^\circ + \cot^2 45^\circ - 2 \cot 30^\circ \cot 45^\circ \cos 100^\circ}$$

as req.

iv) By calculation,
the height of the cliff
is 466 m (to nearest m.)

4/ a) LHS =

$$\frac{1}{1 \times 4} + \frac{1}{4 \times 7} + \dots + \frac{1}{(3k-2)(3k+1)} + \frac{1}{(3k+1)(3k+4)}$$

$$= \frac{k}{3k+1} + \frac{1}{(3k+1)(3k+4)}$$

$$= \frac{k(3k+4) + 1}{(3k+1)(3k+4)}$$

$$= \frac{3k^2 + 4k + 1}{(3k+1)(3k+4)}$$

$$= \frac{(3k+1)(k+1)}{(3k+1)(3k+4)}$$

$$= \frac{k+1}{3k+4}$$

= RHS as req.

b/ Step 1. To prove true for $n=1$
 $2^{3n} - 3^n$ when $n=1$
 $= 2^3 - 3$
 $= 5$ which is divisible by 5
 $\therefore$ true for $n=1$

Step 2. Assume true for $n=k$

i.e. $\frac{2^{3k} - 3^k}{5} = M$
 where M is a whole number

$$\text{i.e. } 2^{3k} - 3^k = 5M$$

$$2^{3k} = 5M + 3^k$$

Step 3 to prove true for $n=k+1$

i.e. $2^{3(k+1)} - 3^{k+1}$ is divisible
 by 5.

$$2^{3(k+1)} - 3^{k+1} = 2^{3k+3} - 3^{k+1}$$

$$= 2^3 \times 2^{3k} - 3 \times 3^k$$

$$= 8 \cdot 2^{3k} - 3 \cdot 3^k$$

$$= 8(5M + 3^k) - 3 \cdot 3^k$$

$$= 40M + 8 \cdot 3^k - 3 \cdot 3^k$$

$$= 40M + 5 \cdot 3^k$$

$$= 5(8M + 3^k)$$

which is divisible by 5,
 $\therefore$ if true for $n=k$ then
 true for $n=k+1$

Step 4 Since true for $n=1$
 then it must be true for $n=1+1=2$
 and if true for $n=2$, then it
 must be true for $n=2+1=3$
 and so on for all positive
 integers n .

c/ (i) $f(2) = 2^3 + 3 \times 2 - 18$

$$= 8 + 6 - 18$$

$$= -4 < 0$$

and $f(2.5) = 2.5^3 + 3 \times 2.5 - 18$

$$= 15.625 + 7.5 - 18$$

$$= 5.125 > 0$$

$\therefore$ a root exists in the interval

$$2 < x < 2.5$$

(ii) let $x = 2.25$

$$f(2.25) = 2.25^3 + 3 \times 2.25 - 18$$

$$= 0.140625 > 0$$

$\therefore$ a smaller interval is
 $2 < x < 2.25$

(iii) Since $f(2.25)$ is closer
 to zero than $f(2)$ then
 the root is closer to 2.25

d/ let $f(x) = x^3 - 6x^2 + 25x - 50$

$$f'(x) = 3x^2 - 12x + 25$$

(if $x_1 = 9.5$) $x_2 = x_1 - \frac{f(x_1)}{f'(x_1)}$

$$x_2 = 9.5 - \frac{f(9.5)}{f'(9.5)}$$

$$f(9.5) = 53.375$$

$$f'(9.5) = 181.75$$

$$x_2 = 9.5 - \frac{53.375}{181.75}$$

$$= 9.711724 \dots$$

5/ a) (i) $\int \frac{\cos x}{\sin x} dx$

$= \ln(\sin x) + C$

(ii) $\int \sin^2 4x dx$

$= \int \frac{1}{2}(1 - \cos 8x) dx$

$= \frac{1}{2} \left(x - \frac{1}{8} \sin 8x \right) + C$

$= \frac{1}{2}x - \frac{1}{16} \sin 8x + C$

b) $\int_2^3 \frac{dx}{\sqrt{x^2-4}}$

$= \left[\ln \left(x + \sqrt{x^2-4} \right) \right]_2^3$

$= \left[\ln(3 + \sqrt{5}) - \ln(2 + 0) \right]$

$= \ln \left(\frac{3 + \sqrt{5}}{2} \right)$

c) $\int_0^\pi (1 + \cos x)^2 dx$

$= \int_0^\pi 1 + 2\cos x + \cos^2 x dx$

$= \int_0^\pi 1 + 2\cos x + \frac{1}{2}(1 + \cos 2x) dx$

$= \int_0^\pi \frac{3}{2} + 2\cos x + \frac{1}{2}\cos 2x dx$

$= \left[\frac{3}{2}x + 2\sin x + \frac{1}{4}\sin 2x \right]_0^\pi$

$= \left(\frac{3\pi}{2} + 2\sin \pi + \frac{1}{4}\sin 2\pi \right) - 0$

$= \frac{3\pi}{2}$

d) $\int \frac{x}{\sqrt{1+x^2}} dx$ $\begin{cases} x = \tan \theta \\ \frac{dx}{d\theta} = \sec^2 \theta \end{cases}$

$= \int \frac{\tan \theta \cdot \sec^2 \theta}{\sqrt{1 + \tan^2 \theta}} d\theta$ $\begin{cases} dx = \sec^2 \theta \cdot d\theta \end{cases}$

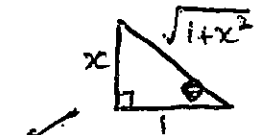
$= \int \frac{\tan \theta \cdot \sec^2 \theta}{\sec \theta} d\theta$

$= \int \frac{\tan \theta \sec^2 \theta}{\sec \theta} d\theta$

$= \int \tan \theta \sec \theta d\theta$

$= \sec \theta + C$

$= \sqrt{1+x^2} + C$



e) $\int_0^3 \frac{3x}{\sqrt{1+x}} dx$ $\begin{cases} u = 1+x \Rightarrow x = u-1 \\ \frac{du}{dx} = 1 \end{cases}$

$= \int_1^4 \frac{3(u-1)}{\sqrt{u}} du$ $du = dx$

$x=3 \quad u=4$

$x=0 \quad u=1$

$= \int_1^4 3u^{1/2} - 3u^{-1/2} du$

$= \left[\frac{3u^{3/2}}{3/2} - \frac{3u^{1/2}}{1/2} \right]_1^4$

$= \left[2u^{3/2} - 6u^{1/2} \right]_1^4$

$= (2 \times 8 - 12) - (2 - 6)$
 $= 8$

6/ a) (i) $y = 2 \tan^{-1} x$

(x=1) $y = 2 \tan^{-1} 1$
 $= \frac{\pi}{2}$

(ii) $y' = \frac{2}{1+x^2}$

when $x=1$ $y' = \frac{2}{2} = 1$

$\therefore$ gradient of normal $= -1$

eqn of normal $\Rightarrow$

~~$y - \frac{\pi}{2} = -1(x-1)$~~

~~$y - \frac{\pi}{2} = -x + 1$~~

~~$x + y - \frac{\pi}{2} - 1 = 0$~~

b) $\int_0^{\sqrt{3}} \frac{4}{9+x^2} dx$

$= \frac{4}{3} \left[\tan^{-1} \frac{x}{3} \right]_0^{\sqrt{3}}$

$= \frac{4}{3} \left(\tan^{-1} \frac{\sqrt{3}}{3} - \tan^{-1} 0 \right)$

$= \frac{4}{3} \left(\frac{\pi}{6} - 0 \right)$

$= \frac{2\pi}{9}$

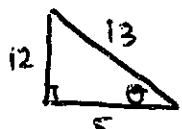
c) $\cos \left(\sin^{-1} \left(-\frac{12}{13} \right) \right)$

Since $\sin^{-1}$ is an odd function

$\sin^{-1} \left(-\frac{12}{13} \right) = -\sin^{-1} \left(\frac{12}{13} \right)$

$\therefore \cos \left(\sin^{-1} \left(-\frac{12}{13} \right) \right) = \cos \left(-\sin^{-1} \left(\frac{12}{13} \right) \right)$
 $= \cos \left(\sin^{-1} \left(\frac{12}{13} \right) \right)$

Since $\cos(-\theta) = \cos \theta$

Now  $\therefore \sin \theta = \frac{12}{13}$
 $\theta = \sin^{-1} \left(\frac{12}{13} \right)$

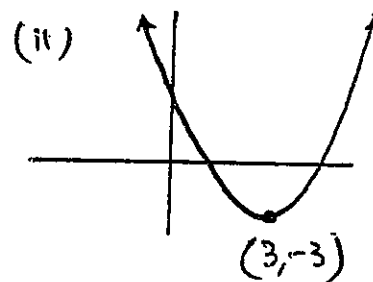
$\therefore \cos \left(\sin^{-1} \left(\frac{12}{13} \right) \right) = \cos \theta$
 $= \frac{5}{13}$

d) $f(x) = x^2 - 6x + 6$

(i) $x = -\frac{b}{2a}$
 $= \frac{6}{2}$

$x = 3 \Rightarrow$ Axis of symmetry

$\therefore$ Vertex $= (3, -3)$



No inverse function since it doesn't pass the horizontal line test $\Rightarrow$ restricted domain $x \geq 3$

(iii) Domain of $f^{-1}(x) \Leftrightarrow$ Range of $f(x)$
 $\therefore x \geq -3$

(iv) $x = y^2 - 6y + 6$
 $x - 6 = y^2 - 6y$
 $x - 6 + 9 = y^2 - 6y + 9$
 $x + 3 = (y - 3)^2$

$$\therefore y-3 = \pm\sqrt{x+3}$$

$$y = 3 \pm \sqrt{x+3}$$

(v) Solve $y = x^2 - 6x + 6$ and $y = x$

$$x = x^2 - 6x + 6$$

$$0 = x^2 - 7x + 6$$

$$0 = (x-6)(x-1)$$

$$\therefore x = 6, 1.$$

7/ a) $\frac{|x-1|-2}{12+x-x^2} \geq 0$

Consider $\frac{|x-1|-2}{(4-x)(3+x)} = 0$

$$\therefore x \neq 4, -3$$

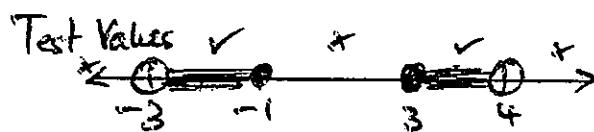
$$|x-1|-2 = 0$$

$$|x-1| = 2$$

$$x-1 = 2, \quad x-1 = -2$$

$$x = 3$$

$$x = -1.$$



$$\therefore -3 < x \leq -1, \quad 3 \leq x < 4$$

b) $f(x) = 2 \cos^{-1}\left(\frac{x}{\sqrt{2}}\right) - \sin^{-1}(1-x^2)$ (iii)

(i) Domain :

$$-1 \leq \frac{x}{\sqrt{2}} \leq 1, \quad -1 \leq 1-x^2 \leq 1$$

$$-\sqrt{2} \leq x \leq \sqrt{2} \quad -2 \leq -x^2 \leq 0$$

$$0 \leq x^2 \leq 2$$

$$-\sqrt{2} \leq x \leq +\sqrt{2}$$

$\therefore$ Domain :

$$0 \leq x \leq \sqrt{2}$$

↑
restriction

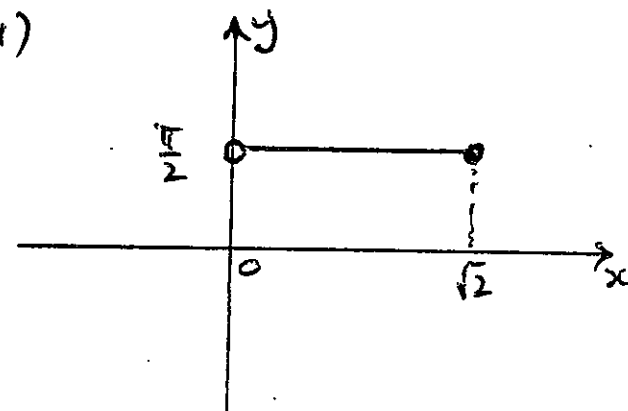
(ii) $f'(x) = \frac{-2}{\sqrt{2}} \cdot \frac{1}{\sqrt{1-\frac{x^2}{2}}} - \frac{-2x}{\sqrt{1-(1-x^2)}}$

$$= \frac{-2}{\sqrt{2-x^2}} + \frac{2x}{\sqrt{1+2x^2-x^4}}$$

$$= \frac{-2}{\sqrt{2-x^2}} + \frac{2x}{\sqrt{2x^2-x^4}}$$

$$= \frac{-2}{\sqrt{2-x^2}} + \frac{2x}{x\sqrt{2-x^2}}$$

$$= 0$$



Since $f'(x) = 0$ $f(x)$ is a constant

when $x = \sqrt{2}$ $y = \frac{\pi}{2}$

$\therefore y = \frac{\pi}{2}$ is the constant

$$c) \frac{\sin\left(\frac{3x}{2}\right)}{2\sin\left(\frac{x}{2}\right)}$$

$$= \frac{\sin\left(x + \frac{x}{2}\right)}{2\sin\left(\frac{x}{2}\right)}$$

$$= \frac{\sin x \cdot \cos \frac{x}{2} + \cos x \cdot \sin \frac{x}{2}}{2\sin \frac{x}{2}}$$

$$= \frac{2\sin \frac{x}{2} \cdot \cos \frac{x}{2} \cdot \cos \frac{x}{2} + \cos x \cdot \sin \frac{x}{2}}{2\sin \frac{x}{2}}$$

$$= \frac{\cancel{\sin \frac{x}{2}} (2\cos^2 \frac{x}{2} + \cos x)}{2\cancel{\sin \frac{x}{2}}}$$

$$= \cos^2 \frac{x}{2} + \frac{1}{2} \cos x$$

$$= \frac{1}{2} (1 + \cos x) + \frac{1}{2} \cos x$$

$$= \frac{1}{2} + \frac{1}{2} \cos x + \frac{1}{2} \cos x$$

$$= \frac{1}{2} + \cos x \text{ as required}$$

d) Let $P(x)$ have a double root at $x = \alpha$

$$\therefore P(x) \Rightarrow x^4 - 4kx^2 + 12 = 0 \quad (1)$$

$$P'(x) = 4x^3 - 8kx$$

$$\therefore P'(\alpha) \Rightarrow 4(\alpha)^3 - 8k\alpha = 0 \quad (2)$$

$$\text{Solving (2): } 4\alpha^3 - 8k\alpha = 0$$

$$4\alpha(\alpha^2 - 2k) = 0$$

$$\therefore \alpha = 0, \pm\sqrt{2k}$$

Clearly $\alpha = 0$ is not a solution as $P(0) \neq 0$.

$$P(\sqrt{2k}) = 4k^2 - 8k^2 + 12 = 0$$

$$-4k^2 + 12 = 0$$

$$\frac{-4k^2}{4} + \frac{12}{4} = 0$$

$$\therefore k = \pm\sqrt{3}$$

$P(-\sqrt{2k})$ gives the same values of k

$$\therefore k = \pm\sqrt{3}$$